Part IV

Conclusion

14

Conclusions

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14.1 Introduction

If you have read each of the contributions in this book prior to starting on the conclusions you will already recognise the fact that there are some clear themes running through the chapters. In Fig. 14.1 we have attempted to group them under five separate headings, although they are inextricably linked, as we will see in the discussion that follows. Whilst the number of companies and regulatory authorities who have contributed to this book is not large enough to form the basis for a statistically-based piece of research, contributors have been deliberately drawn from large and small businesses in both the developed and developing markets, and from four continents across the globe. What they have in common is hands-on experience of real HACCP systems. They are not theorists but practitioners. This emphasis on practical experience is the most exciting thing about this book and the lessons that we can extract from that experience.

It is interesting to note that 40 years after the HACCP concept was first conceived and 30 years after it was first published, there are so many controversial issues still associated with its effective implementation. Is this because the concept is flawed in some way? Barnes mentions the codification of HACCP in 1993 through Codex Alimentarius Commission (CAC). Through the publication of an international HACCP standard and accompanying guidelines, there is a standard international reference to the principles and implementation of HACCP systems. However, it is clear that there remain not only many problems but also many misconceptions associated with the use of HACCP methodology. Many countries are represented on the CAC, and a few of its members are contributors to this book. It would be interesting to evaluate how

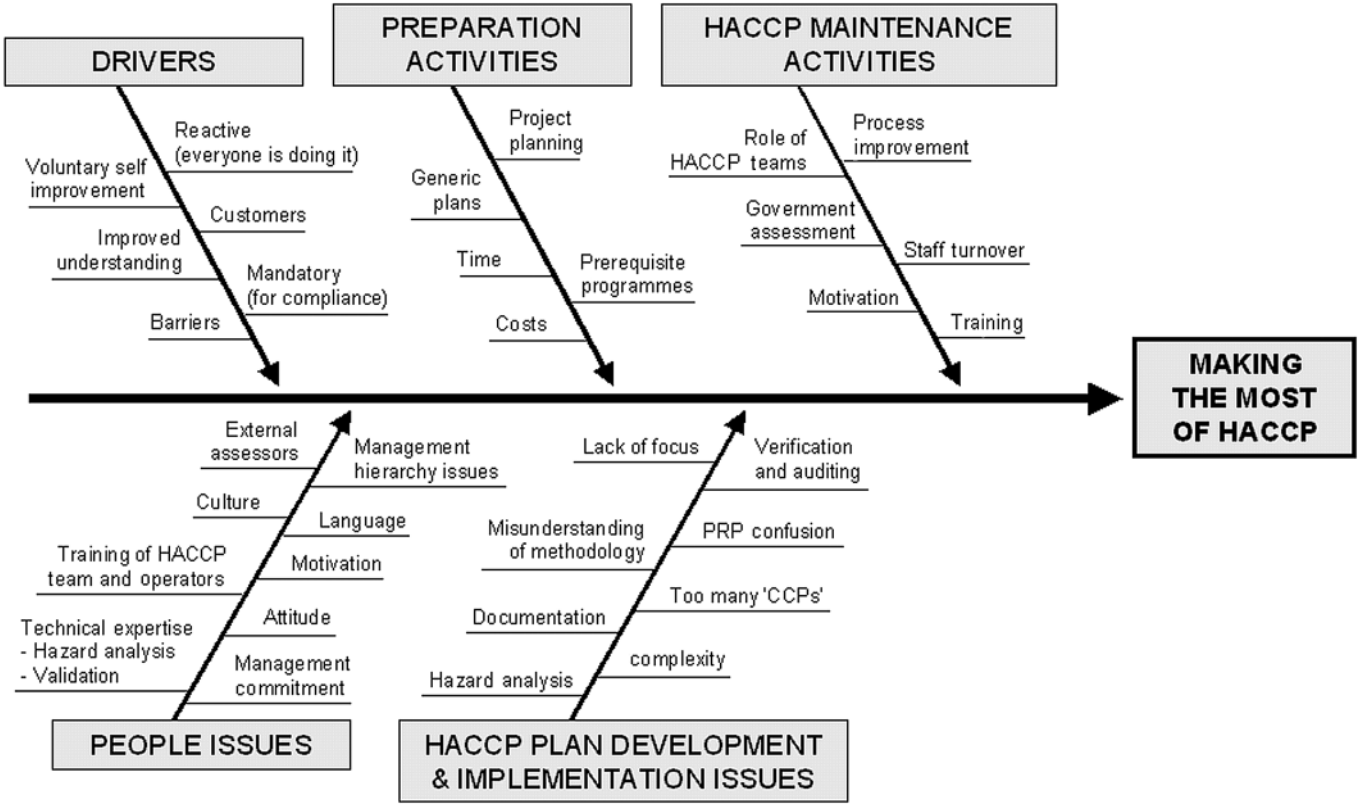


Fig. 14.1 Common themes and issues observed in implementation.

many of the other members have actually participated in a HACCP team. How many years might it be since they worked in a foodservice kitchen or in a food plant, if they ever did? How representative, therefore, of the users of HACCP is the body responsible for drawing up the rules? A controversial point to make perhaps but it highlights the importance of reflecting on the experience of those who have hands-on experience of putting principles into practice and the problems they encounter. How can these problems be addressed? We will attempt to offer some suggestions as we examine each of the themes identified. In many cases we can offer pointers on avoiding common pitfalls, though some issues require action at government level. Hopefully readers will be drawn from both camps: those who are yet to begin their HACCP programmes or who want to improve on what they have, and those who are responsible for regulation and setting the standards. The rest of this chapter looks at each of the five themes identified in Fig. 14.1 and what can be learnt in each case about problems and solutions in successful HACCP implementation.

14.2 Learning 1: drivers for HACCP

Typically the motivation for putting in a HACCP system tends to come from:

- customer pressure (Kane, Suwanrangsi, Route)
- regulatory requirement (Marthi, Taylor, Suwanrangsi)
- desire for self improvement (Killen, Lee, McAloon).

Taylor notes that many SMEs are ignorant of food safety issues and the management systems required to deal with them, and that the only pressure to adopt HACCP comes from legislation. This view is echoed by Suwanrangsi, and both authors point to a lack of real motivation even with legal compliance as a driver. Taylor suggests that this is because the risk of prosecution for food hygiene offences is low in many countries. However, the consumer expects safe food, whoever produces it, in whatever size business and in whichever country. In India, Marthi points out that there is no perception of risk by the consumer and therefore no desire to improve food safety from their end of the supply chain. With no perception of risk, it is difficult for industry to sell a food safety management concept to consumers which in turn reduces the commercial rationale for improvements in relation to cost. India, like a number of other countries in Asia and South America, has a history of over-population and lack of food. It took trade problems with seafood exports to the USA and Europe to prompt the Indian government to mandate for HACCP in that sector. As a result India now acts as a spokesperson for developing countries in Codex policy-making on HACCP.

In Thailand there is also a large seafood export industry but Suwanrangsi makes clear that the industry still did not embrace HACCP at first. They relied too much initially on external pressure, customer advice and generic models which resulted in ineffective systems. The real change in momentum came when HACCP was seen as essential for competitive success in improving both safety

and quality. In Thailand businesses needed external pressure to influence attitude change, but they needed to see and understand the benefits in order to take the initiative in HACCP implementation. The importance of companies taking the initiative for themselves is emphasised by other contributors. In New Zealand, Lee describes an unregulated voluntary approach, noting that it results in a better 'buy-in' than mandatory approaches. However, it can be slow to happen (unless driven by customers) and it can be inconsistent in that no single standard is adopted.

Barnes, Taylor, McEachern and Route all noted the slow uptake of HACCP by SMEs. In Route's case it was customer pressure that got the momentum going. The typical senior management view was that they had neither time nor money for implementation of an activity that they perceived as irrelevant. Once HACCP became a commercial necessity attitudes changed. Route concludes that this driver is essential since, in small businesses, very little can be achieved without key management commitment, although he notes that HACCP implementation can take place without the active involvement of senior management.

Why then do SMEs not see HACCP as a real benefit? Why do they shy away from it? Barnes rightly points to the WHO consultation (WHO, 1999) on small but also less developed businesses (SLDBs). Several authors (McEachern, Taylor, Barnes) point to concerns about costs, time, complexity, perception of risk, and lack of understanding of commercial rationale. They even suggest that this reluctance may, in part, be due to the initial uptake of HACCP by larger manufacturers (Barnes). As pioneers in HACCP implementation, some large manufacturers developed large, unwieldy and complex HACCP systems. This over-bureaucratic approach was not initially acknowledged but instead seen as a sign of their superior knowledge. With hindsight it is possible to see that lack of understanding, the need for political balance, the divergence of approaches between industry-owned, voluntary HACCP systems, and government-sponsored, mandatory HACCP systems, all contributed in some way towards the development of a 'large industry' HACCP model that did not take account of SME capabilities.

In summary, it seems that the best HACCP systems are developed by businesses that are self driven. The prompt may come from the regulators or customers or an enlightened individual within the company. Key levers to support the adoption of HACCP will include:

- Acknowledgement of the continuing rising trends in foodborne illness – HACCP needs to be diligently adopted throughout the industry if this is to be permanently reversed.
- The need for businesses to see HACCP for what it is – as a 'minimal' system allowing businesses to understand and focus on the critical safety areas, and be better able to use resources effectively.
- The business benefits of HACCP in terms of customer reassurance, competitive advantage, process improvement, need to be better communicated and real examples publicised where possible.

- The need for HACCP to be seen as a tool for 'empowerment' allowing businesses to devolve control of CCPs in an organised and safe way – involving employees, who are the ones who make safe food in any case.
- In the future it is likely that HACCP will become outcome oriented and will be more clearly linked to a reduction in the risk of consumer foodborne disease.

Government involvement in particular can be helpful with SLDBs. McEachern relates how contact with government inspection staff helped to sell the benefits to SMEs in Canada. Suwanrangsi also suggests that a proactive and close collaboration between industry and government can really help in successful implementation of HACCP through promotion of HACCP and its benefits, advice, training and other resources, and financial support. A similarly close collaboration between industry and government also offers the best opportunity for an effective assessment of HACCP implementation and improvements to HACCP systems (see Chapter 15).

In reading the regulatory chapters one is struck by the changing roles of regulators. McEachern describes the difficulties in moving from a traditional inspection to a verification role although he is convinced that the latter provides a greater assurance of food safety. The difficulties arise partly in getting regulatory assessors with the skills to conduct a verification audit (acknowledged by Lee, Barnes, Taylor and Suwanrangsi). They also come from persuading food businesses that they need to change their expectation that government inspectors will come in with a tick list and simply tell them what to do. Under a HACCP system, the food business is responsible for making safe food and the regulator is responsible for verifying that the operator is capable of doing so. Lee describes a similar shift in New Zealand to a model where the government sets food safety standards, the food industry accepts responsibility for food safety management through HACCP systems, and independent auditors verify that industry has met its responsibilities.

One of the major challenges facing governments is the level of support they provide to industry. Governments have responded in differing ways to the needs of their food industries in HACCP implementation, but have tended to increase the level of support provided as they have become more aware of industry needs. Lee describes the New Zealand government providing generic HACCP plans backed up by information on key hazards and process steps, help in setting up a national microbiological database for the export meat sector, and other technical support for the food industry. Suwanrangsi points to the similar development of generic HACCP plans and advice on hazards in Thailand, and Marthi emphasises the need for government involvement in providing relevant information on hazards for Indian food processors. The New Zealand government has also worked with the industry to develop HACCP competency qualifications for both industry and regulatory personnel. Barnes and Taylor note the UK government developing resource centres and training materials as well as funding HACCP training and consultancy, developments also described

by Suwanrangsi in support by the Thai government for key export sectors such as fish. This growth in support emphasises in particular the importance of people as a key resource in HACCP implementation, their education and training, and their development of a true understanding of the HACCP method – what it is and what it is not. These 'people' issues are discussed in the next section.

14.3 Learning 2: people

In drawing out the issues raised by the contributors, overwhelmingly the biggest number is related to people – their lack of training, poor-quality training, attitudinal problems, technical knowledge and much more. This was not just a question of needing more training, but a much deeper issue which we will attempt to analyse. It underpins all other activities, including the preparation process for HACCP implementation.

14.3.1 Motivation, attitude and cultural issues

The purpose of training can be described as twofold (Wallace):

- to develop awareness and to motivate
- to develop technical knowledge (in this case in HACCP planning and implementation) enabling full participation in development and implementation of the system.

Failure to achieve either of these aims properly could lead to failure of the HACCP system from the very beginning. As mentioned earlier, it is harder to develop a HACCP system in the absence of management commitment. In fact, it is harder to have a reliable food safety and quality programme of any sort. Prerequisite hygiene programmes are an essential partner to the HACCP system and management commitment is needed for these to be embedded in the heart of the business.

It was interesting to read McAloon's reports of setting bonus incentives for managers based on continuing good performance in food safety management, suggesting that continuing commitment to food safety may need incentives. It may be there when people have spare time or are involved in a specific project (e.g. HACCP plan development), but it cannot be assumed to be present on a routine basis. Another recommendation by McAloon was that, to get food safety into the culture of the company, it is a good idea to assign responsibility for HACCP implementation to a senior company person. This can be seen as a 'positive act' as opposed to stating, as many companies do, that 'all senior managers are responsible'. There must be real leadership at top level whatever size the business. Such an approach also provides the HACCP team with a senior level 'champion' who can ensure that their interests are listened to in the right quarters.

Management commitment is cited as being an essential starting point in most textbooks and journals on this subject, and we have highlighted a couple of ways

of achieving such commitment. What about the lower levels in the organisation, the supervisors and operatives? Often they can be cynical, with a 'seen it all before' attitude and a reluctance to adopt new work practices. Sometimes this attitude has developed as a result of a prescriptive management culture – management being unwilling to accept the views of more junior employees. Both Killen (UK) and Route (China) mentioned this and offered some ideas for securing employee commitment and participation. Killen suggests that, on occasions, the managers and operatives can be separated for any hazard analysis and problem-solving work. This enables a freedom of thought and speech during team sessions and gets the best out of both groups. It requires cultural awareness by the HACCP team leader but, gradually, as both groups see the value of the other, they can come back together. In Route's example there were real cultural differences between East and West to be acknowledged and utilised to achieve the end result. When working in a foreign culture it is essential to understand those differences when beginning a business relationship. 'Challenging' as a way of problem solving is not normal in the Chinese work ethic and therefore tact and understanding was needed in order to work within this culture. Validation and verification activities, for example, were viewed initially with suspicion as questioning employee competence, and care was needed in communicating what was needed. However, it seems, in this example, that employees demonstrated a genuine commitment and enthusiasm once this initial concern was overcome, amply repaying the effort involved.

In some cases the company culture is identified as a problem for managers and operatives alike. Suwanrangsi describes a reactive culture where they wait for the official inspection before doing anything. McEachern describes a paternalistic culture where food safety is seen as a government responsibility. In these cases government departments were active in trying to act as a catalyst for culture change, e.g. through briefing sessions for senior management. The point was made that it was better to start the change in attitude prior to starting any work on the HACCP programme.

At operator level, several chapters (Rudge, McAloon, Route, Killen) mentioned the need for their active involvement at an early stage. McAloon points out that the line workers, not the managers or the Quality Department, make safe food – so involve them. Managers in the Quality Department are, of course, critical, both in the development stage and in a verification role once the plan is implemented. However, ownership has to transfer at some stage – hence the importance of involving line staff sooner rather than later. There is broad agreement that employees should be involved in writing the HACCP plan but differences appear when considering at what stage. Rudge describes how a number of employees were brought together as a team to draw the Process Flow Diagram and carry out a Hazard Analysis but that he and a colleague were responsible for determining CCPs. This approach might alarm the purists (i.e. a HACCP team needs to consist of four to six people, ideally, and should be multi-functional) but there may be good reasons for taking an alternative approach. In this case there were language issues at the Polish plant in question, but anyone who has observed and participated

in a number of HACCP teams will know that discussion can be lively, technical and, on occasions, acrimonious. For the less technical members of a HACCP team this might actually prove to be demotivating and confusing. Having them review the output might be a better approach. Killen notes a similar approach in describing how the HACCP team leader, in undertaking a preliminary Hazard Analysis, can help in preparation for team meetings. It may be best to involve line staff most where they have the most to contribute, particularly in such areas as process analysis, where they may well have the most hands-on experience, and in designing CCP procedures and documentation, where they can reflect the practical issues involved in CCP monitoring.

14.3.2 HACCP Teams

A discussion of HACCP Teams leads nicely on from the above. There is some broad agreement amongst the contributors in this area:

- HACCP is best done as a team (five or six people at the maximum is said by Killen better to enable full discussion and contribution from all members).
- The HACCP Team needs to be multi-functional and have a knowledge that has depth and breadth across the enterprise (McAloon, Taylor, Killen and Route).
- SMEs have limited resources but this can be turned to advantage in that the team members tend to be really close to the process and have in-depth knowledge (Taylor, Route). A small team can turn out to be flexible and dedicated to the job they have been asked to do, and the individuals selected can act as a catalyst for change. Ownership of the HACCP plan by those who will then have to implement it results in a stronger system.
- Challenge within the HACCP team is important as the most senior person, or the person with the most technical-sounding job title, does not always have all the ideas or all the answers. This environment of challenge culture is worth developing if the HACCP system is to be as good as possible.
- Using operator level people in the team, even for just one part of the process, can be a great benefit. They go on to make excellent CCP monitors, act as HACCP champions on the shop floor, can help to design documentation and generally help to improve the process.

14.3.3 Use of consultants, external support, information sources

It is well debated and documented that SMEs rarely have the skills, knowledge or resources needed to develop and implement a HACCP programme. It was gratifying to read that skills shortage at the start of HACCP implementation is not just related to the size of a business but is also a problem for larger organisations (McAloon, Rudge). This is where the use of internal or external expert consultants can be a good way of getting started. Several examples are given, some fairly innovative, of how to fill the knowledge gap.

Use of expert external consultants

McAloon related how this was done within Cargill (Fig. 14.2). Whilst this is what was done for HACCP, the same approach could be used for any new programme.

Large companies do not necessarily have company experts already within their employ and the employee who knows more than the others is wise to admit that they do not know everything. An external expert was utilised at Cargill for a period of six months with the aim of developing in-house expertise that could then be cascaded throughout the business.

The experience of SME's use of external consultants was more mixed (Taylor). Problems included the variable quality of consultant advice, the unfamiliarity of SMEs with the key issues on which advice was needed, and the consultant's unfamiliarity with the business's particular processes and requirements. Taylor and Barnes suggest that these problems can be overcome by better initial training for SMEs, allowing them to do more of the groundwork before approaching a consultant, selecting an approved consultant, and using consultants to develop and help validate a HACCP system rather than employing them right at the start.

Use of internal experts

In Rudge's case, a similar approach was taken at the next level down from where McAloon started. In this example Rudge himself became the vehicle for ensuring that what was known at a business headquarters level was transferred to the operating plants. He facilitated visits between sister plants as a way of sharing this knowledge. Marthi describes a similar scenario using both external support and staff from the company's research centre to disseminate knowledge and understanding of HACCP principles at senior management level, and then

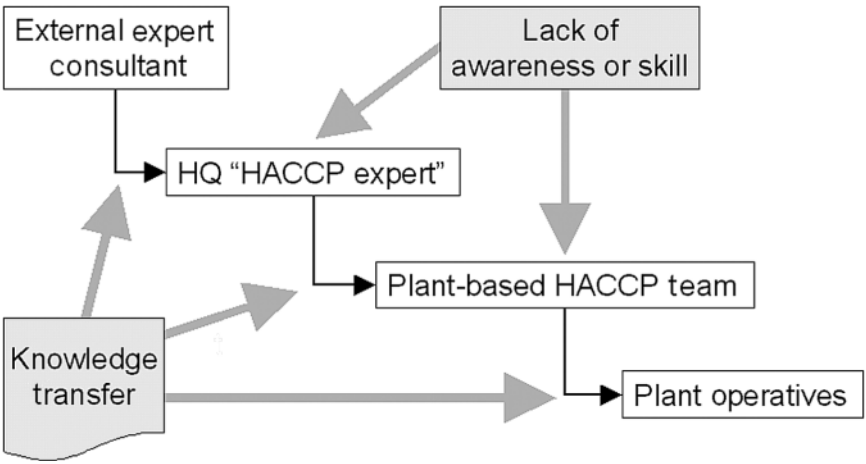


Fig. 14.2 Skills and knowledge transfer in Cargill.

within the management teams at individual factories. The disadvantage of this process of cascading knowledge from the centre was that experts were not always familiar with the individual businesses and had to learn. Rudge emphasises how useful the development of the Process Flow Diagram is, at this point, to aid understanding of a particular operation and its particular characteristics.

Customers, regulators, published work, competitors

With some innovative thinking information can be found fairly cheaply. Some examples and solutions are provided in the chapters. As an example customer audits can be seen as free consultancy, particularly in relation to prerequisite systems governing hygiene and GMP. If auditors are experienced, through seeing many different HACCP systems, such audits can be very helpful (Kane). However, care needs to be taken as sometimes the auditor has limited practical experience and understanding, and the business has to challenge recommendations made (Route). Figure 14.3 illustrates some of the sources of knowledge and skills transfer.

Regulatory inspectors and assistance

We have heard already how the role of the regulators is changing and that, for many start-up businesses and SMEs, rightly or wrongly this is the main source of information. As with auditors, however, the quality of such information and advice depends on the training, experience and understanding of regulatory staff. Suwanrangsi notes a dialogue between a company and regulator which resulted in the latter reassessing the need for a particular CCP.

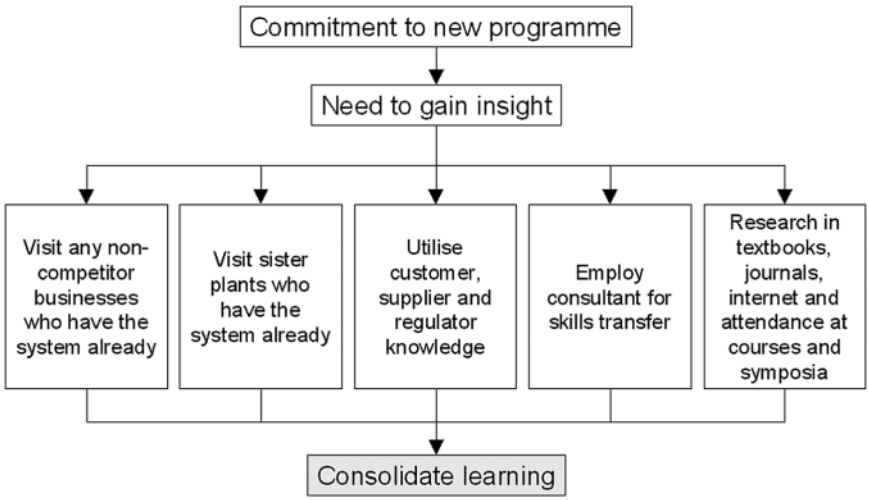


Fig. 14.3 Sources of knowledge and skills transfer.

Published work
Route mentions the internet but, in addition, there is an increasing array of publications in which to discover information. A significant amount of information is available if a company is proactive and looks for it. One afternoon spent identifying relevant published information prior to begining the HACCP plan would, in the majority of cases, be time very well spent. Taylor describes the development of resource centres to vet and pool relevant information so that companies can access reliable information more easily.

Competitors
Taylor describes how, in the UK, the HACCP initiative for butchers enabled them to put aside their competitiveness and to share ideas, conduct visits to each other's premises and to exchange good practices. In this instance the whole meat retail sector was compromised by a serious outbreak of E.coli food poisoning, making cooperation between businesses a necessity in restoring consumer confidence.

14.3.4 Language issues

A couple of contributors shared important lessons related to language and communication. Many of the published materials and experienced HACCP expert consultants are Western in origin and, whilst materials can be translated (although even this can be difficult if non-technical translators are used), the expert consultants and trainers are rarely sufficiently fluent in other languages to enable effective knowledge transfer. Some solutions to overcoming language issues are:

- Translate the Process Flow Diagrams and use them for communication (Rudge).
- Use a bilingual expert in HACCP rather than a translator wherever possible (Rudge).
- Allow extra time for training and project work as everything takes twice as long (Rudge, Route).
- Check the competence of the translator carefully as sense and meaning can easily be lost (Rudge). If possible use in-house people, though be careful if they are non-technical, e.g. marketing or sales personnel (Suwanrangsi).
- If training through an interpreter, find other means to check understanding as it will be difficult (and frustrating!) trying to do this through the interpreter (Rudge). Other means might include some form of validating multiple-choice tests.
- If there is someone locally who is fluent in English then train them as a trainer rather than have a third party expert try to train the whole group through an interpreter.

Appropriate language is important for communication through the business. Suwanrangsi makes the point, which I am sure many practitioners share, that

company procedures must be in terms that workers understand at their own level. Whilst export businesses may need to present HACCP information to overseas customers in the customer's own language (in many cases this may well be English), at plant level the information needs to be in the local language. It must also be presented in simple terms that will be easily understood. Rudge actually recommends not sharing all information with operators but providing specific information that is relevant to them if it is to be inculcated throughout the organisation. In Rudge's case laminated cards were provided with the information that was really needed by operators to do their job as opposed to giving them a confusing amount of 'background' information.

14.3.5 Lack of expertise and technical knowledge – educational issues

Whilst most people can understand the concept of HACCP, not everyone has the technical knowledge and expertise to use HACCP principles for effective food safety management. HACCP is a logical and common sense approach to control potential foodborne hazards. The concept can be, and in some countries including the UK is, explained to schoolchildren. With a talented trainer even the most hardened cynic can be made to see that it is based on a sound principle. There remain, however, a number of really important educational rather than training issues to resolve if it is to be made possible for all food businesses to adopt it.

Misinterpretation of methodology

Whilst the concept is straightforward, to be able to perform a Hazard Analysis and determine criticality takes a lot more knowledge than can be absorbed on a two-day training course. Typically, as quoted from Taylor (although this is not solely a problem confined to SMEs), there is:

- difficulty in prioritising risks, for example, physical vs. microbiological risks
- lack of appreciation of which pathogens are relevant (all the bugs that can be found in the book end up appearing in the HACCP plan)
- inability to understand the epidemiology and therefore which pathogens are of concern.

Lack of availability of expert advice and training

There are over 600,000 food businesses in the UK alone, placing great pressure on resources in terms of training, education and expert support either from government or from third-party experts. Taylor says that the single most important factor in getting a HACCP system implemented was the access to an experienced specialist. In Thailand this expertise is provided by the government, at least for the seafood industry, though due to the conflict of interest (government is also the assessment body) regulators have to be fairly general in the advice they give. The inability to make informed and knowledgeable decisions results in confusion, overload and dilution of control. This is one of the biggest causes of system failure.

Calibration and standardisation of training providers
Both Rudge and Wallace mention the difficulties in locating a reputable source of HACCP training. There is at present no recognised international body that can certify training centres, trainees and their materials. There are various national organisations but comparison of training providers on an international basis is not possible. We offer no immediate solution and can recommend only that qualifications are thoroughly assessed for both proven experience of HACCP and Hazard Analysis and training skills, as opposed to lecture and presentation skills. It remains a rarity to find both in the same individual.

14.3.6 Problems related to HACCP training and education

This is not highlighted as a major barrier or difficulty by any of the contributors, yet many of them make helpful recommendations for ensuring that this is done properly. In Wallace's chapter there is guidance on learning styles and classroom layout with a list of reasons why training may be ineffective and what can be done to make it successful (Table 14.1). However, much of the knowledge and skills required for HACCP implementation cannot be acquired through a single short training intervention. Knowledge of hazards and their control, along with good hygiene and manufacturing practice, takes years to acquire. Many businesses would need to buy this expertise in either through new recruits or the use of consultants. Barnes suggests the need to move to 'second-generation' HACCP systems which are less prescriptive and more suitable to particular kinds and scales of food business. This shift may involve greater emphasis on strengthening the prerequisite good hygiene practice, and the use of generic HACCP plans which can easily be adapted to certain types of food business. Although he writes from the perspective of a larger manufacturer, McAloon points out that strengthening prerequisite systems can make HACCP planning and implementation much simpler, a point endorsed by Taylor in her experience of working with SMEs. Increased support reinforced by external inspection by government, says Barnes, will not compensate for inadequate food safety management – nor should it try to. The role of government in providing a knowledge base to fill the educational void in both large and small business needs, however, to be re-examined. This may come with the adoption of Risk Analysis Principles particularly as HACCP becomes outcome oriented (see Chapter 15).

14.4 Learning 3: preparation for HACCP

The time spent preparing to develop a HACCP system is time well spent. Almost all of the contributors have some lessons to offer in this area. As a general observation, businesses often tend to launch themselves straight into the drawing up of process flow diagrams and having a go at a hazard analysis. Lack of preparation leads to confusion, over-complex and ineffective HACCP systems.

Table 14.1 Reasons for ineffective training

Reason	Possible solution
Lack of motivation	Ensure that the trainees are briefed prior to undertaking the training. They need to know what their role is in relation to the topic and why they are being sent on the course. When they are on the course it should be made relevant to them. Both Rudge (house analogy) and McAloon stress that they should not just be told what to do but why they should do it.
Lack of understanding of the type of training needed	It is necessary to understand the task to be achieved following the training. For example, if a trainee is charged with leading a HACCP programme then a half-day 'Introduction to HACCP' course will not produce an expert who is up to the job. The other aspect here is the learning approach to take. Considering the behavioural approach, Rudge uses the car analogy which is the correct approach when training employees to do such things as washing their hands. But it is probably not appropriate when training CCP monitors when you need to create a non-thinking 'auto-pilot' style of operative. In fact this is cited by Barnes as being one of the causes of failure at the CCP monitoring/corrective action stage. A cognitive approach combined with experiential learning is likely to be more appropriate. As an example, Kerry took operatives to a sister plant to watch, participate and learn. Suranwangsi mentions that training should be kept simple but that operatives should be involved in order to aid understanding. Operatives are often not used to taking decisions or using their initiative, a behavioural approach to training would not encourage this. Killen described an experiential approach as being potentially problematic, so we learn that there may be pitfalls to encounter. He points out that giving training 'on-the-job' can sometimes reinforce bad habits if not carefully managed and that a suitable solution is to back this up with classroom training in addition to assessing competence on the job.
Training the wrong people or training at the wrong time	A number of considerations can be highlighted from the experience of the editor such as expecting someone like a maintenance technician who is new to the food industry to be able readily to grasp the principles of HACCP (Wallace also recommends ensuring that trainees have been trained in hygiene beforehand). Another mistake would be to train a HACCP team who then do not begin the project until 12 months later.
Lack of follow-up	Reinforcement after training is a well-documented requirement if the learning is to be cemented in the minds of the trainees. Wallace described the Cadbury Schweppes approach of following the introductory course with a facilitated workshop.

Table 14.1 Continued

Reason	Possible solution
Poor-quality training	This can encompass a lack of understanding on the part of the trainer, a lack of ability to transfer knowledge effectively, a poor scope and quality of training materials and inappropriate learning styles. These will all contribute to a poor outcome. There are advantages in using in-house trainers. McAloon supports this by pointing out that courses can be tailored to suit the group of trainees and more people can be trained. As a result of developing in-house trainers you also create HACCP champions and, as a greater level of expertise is required to run a training course, the overall skill base within the company reaches a higher level. As mentioned in the text, there is no recognised standard for HACCP training certification and, therefore, when using external trainers it can be difficult to assess the quality of trainer beforehand. Businesses have to choose between the good and the bad – and cost is not necessarily a good indicator.
Coverage – not training enough people	This can be a problem, particularly for SMEs who cannot afford to send many (if any) people on external training courses. The benefit to having in-house trainers is that more people can be trained.

14.4.1 Prerequisites for HACCP

Which comes first in terms of development – prerequisite (hygiene) programmes (PRPs) or HACCP? Lee argues that the prerequisites should always come first as, otherwise, HACCP development could be hindered. Suwanrangsi, Kane, Taylor, Rudge, Barnes and McAloon all agree. The most successful implementation of HACCP is done within an environment of well managed prerequisite hygiene programmes. This is also the recommendation of Codex (Codex, 1997a, 1997b). Can you be working on HACCP and PRP development at the same time? In practice this is generally what many businesses do, but Suwanrangsi suggests that it is advisable to set up separate project teams, both of which might report to the HACCP Team leader. This approach allows the business to focus resources and skills more effectively. Rudge advocates the use of baseline audits as a good way of assessing the quality of PRPs. This can be done in-house if skills and knowledge are available, or by a reputable third party. In the vast majority of cases PRPs will already be in place, but, as McAloon points out, they just may not be termed as such, formally managed or standardised.

14.4.2 Cost vs. fiscal benefits

Business managers are usually interested in how much HACCP will cost to develop and implement. Technical and quality managers will quote, from

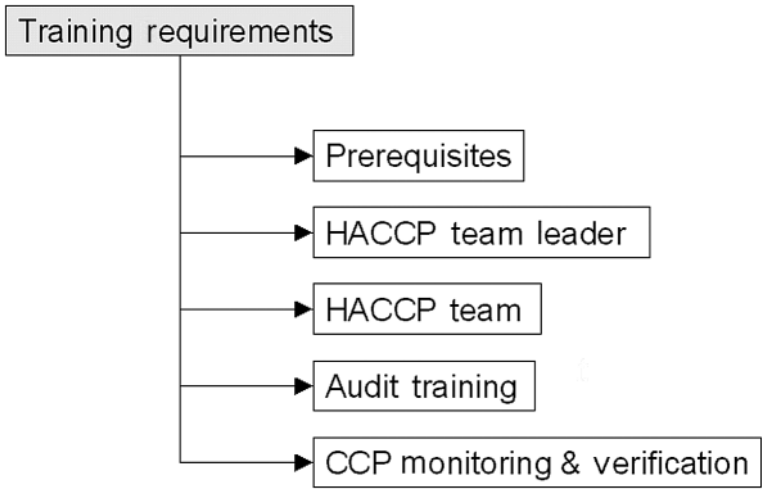


Fig. 14.4 HACCP training requirements.

various publications, the benefits that the company will achieve once HACCP is implemented, but examples are rarely given. Companies are not very good at, or do not have the accounting systems for, tracking cost benefits and attributing them directly to HACCP or to any other quality initiative. In this book we have two very good examples. McAloon puts a figure alongside some real benefits and Marthi mentions an increase in productivity, efficiency and product safety (decrease in Listeria counts). He reports less downtime in production and a reduction of over 10% in the cost of poor-quality product to the business after only six months. What a great set of examples to take to senior management!

It is better to acknowledge that there will need to be an investment in order to reap such rewards – how much is hard to estimate. As McAloon points out, it depends totally on how well the business was being run prior to HACCP. Rudge and Marthi say the same thing. It also depends on what resources are available within the organisation already and what training is needed. Suwanrangsi says that many companies underestimate the amount of training that is needed and the areas that it must cover (Fig. 14.4).

A baseline audit of training requirements, researching the available resources and then drafting a costed training plan will help to avoid surprises later on. As Rudge mentions, it is better to overestimate in terms of budget requirements.

14.4.3 Time issues

After wanting to know how much it will cost, the second question is usually ‘how long will it take?’. The time required depends on a number of factors:

- knowing the starting point and how much there is to do
- what resources are available – knowledge, people, time and money

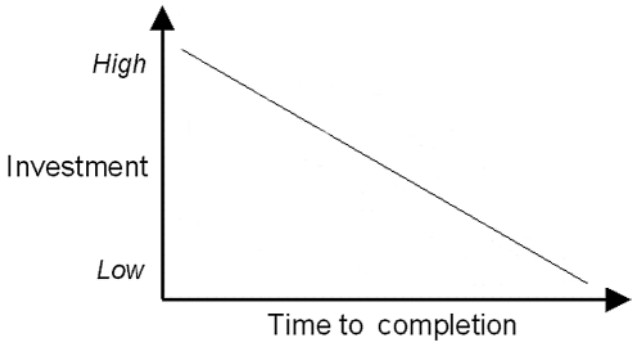


Fig. 14.5 Cost/time ratio diagram.

- the size of the business in terms of sites and production lines
- the complexity of the business (for instance, it will take longer to develop a plan for a chilled Pizza company than for a frozen sweetcorn line)
- the scope to be undertaken in the HACCP study – microbiological, chemical and physical.

To some degree there is a relationship/trade off between the level of investment and the speed with which a HACCP system can be implemented (Fig. 14.5).

The contributors demonstrate the variation. Route quotes two weeks following customer pressure – and for a simple dry blending process this seemed to be possible. Marthi mentions eighteen months as one example. A valuable lesson shared by McAloon is that the timescale for development of GMPs or PRPs can be a lot longer. GMPs cover an entire facility and often involve the implementation of a behavioural change by operatives and management that cannot be rushed. The main lesson is not to try to do too much too fast – be realistic in your expectations.

14.4.4 Generic HACCP plans

Generic HACCP plans are examples of HACCP plans developed for a particular food commodity or process that may be used for guidance (WHO, 1999). The use of these plans is still hotly debated in some circles but several of our contributors reported the use of generic models and their potential benefits. Generic HACCP plans are developed and used in a number of ways:

- *As an industry template*. Here a model HACCP plan is developed, often with input from members of a trade association, together with brought-in experts.
- *In-company generic plans*. This could be, for example, for fast-food restaurant chains. A company may develop one model HACCP plan because its plants or restaurants are making the same range of products using the same prescribed processes across a number of locations.

- *Government/academic prepared models*. A number of institutions have prepared HACCP plans that can be used by businesses as a start point. A number of generic HACCP plans are now available on the internet.

The important thing to remember (Suwanrangsi, Rudge, also WHO, 1999) is that the generic plans must be localised to reflect what is really happening in relation to a specific food or process. In Thailand, government inspectors assessed the amount of reworking of generic HACCP models and used this as a measure of business commitment to and understanding of effective HACCP implementation. In Rudge's case the plan from a sister company was used as the start point and subsequently adapted, saving a lot of time and effort. Barnes also mentions the SLDB report (WHO, 1999) that suggests that generic plans may form a useful starting point. Alternatively, they can be used once the business HACCP team has completed its HACCP plan as a check that their output is not inconsistent with expectations. A potential drawback is that, whilst the use of generic plans may save time and effort, true understanding and ownership must still be developed through developing a specific plan for the business from the start. If a generic plan is implemented without adaptation to the product/process in question, then the local HACCP team will not have been through the detail of the hazard analysis, etc. sufficiently to have ensured that the generic plan was appropriate, and would certainly not have taken real 'ownership' of the plan. Commitment to such a plan is therefore likely to be questionable. In reality, the availability of appropriate generic plans is worth researching and well worth considering when starting a programme, but the local HACCP team must develop its own understanding of HACCP principles, adapt the generic model to the needs of the business and take ownership of its own HACCP plan.

14.4.5 Project planning

Once the baseline audits, availability of generic plans, scope of the HACCP system, etc., has been determined, the business will benefit from the preparation of a full project plan. This may not be highly detailed but it will help to track the progress being made and will prompt the HACCP team to structure the HACCP system and to prioritise tasks. There are a few examples of this kind of planning, all from the larger enterprises. Two conclusions can be drawn:

- A large company is at a disadvantage over an SME given that the size of the task is often much larger. Cargill, Heinz, Unilever and Kerry have many plants worldwide that have, as do other multinationals, multiple production lines within each plant.
- Management teams in larger organisations are used to a more structured approach to project management. This is, perhaps, something that SMEs can learn from.

Even a large business does not have infinite resources in terms of skilled, knowledgeable people or capital to spend on GMP improvements or HACCP

implementation. They are commercial enterprises just like SMEs and they have to prioritise. In the case of Cargill this was done by a risk assessment of the business. This was logical and made the task more manageable. A key lesson here is not to do more than can be managed. Cargill focused on training and supporting their priority plants in HACCP implementation before moving on to other plants. Killen notes that Heinz had to decide whether to perform Hazard Analysis on multiple production lines simultaneously, and thus quickly, or sequentially. They chose the latter option because, like Cargill, it did not overstretch their resources and they benefited from accrued knowledge as they moved from line to line. In contrast Barnes mentions the option of grouping HACCP plans by product and process in order to reduce complexity. Grouping HACCP plans by product and process does provide significant resource savings where, for example, a large number of products, differing only by raw materials content, are manufactured on the same process line.

14.4.6 Communication within the business

Not many contributors mention this but it presents a problem for many companies. Again we see SMEs at an advantage in terms of communication within their business simply on a matter of scale. McAloon reports on the difficulty and challenge in communicating food safety within a plant that has over a thousand workers. Some solutions are offered that may be helpful in small and large businesses alike:

- proactively set up a formal system for communication, e.g. a food safety committee
- publish newsletters with information on anything from HACCP development and progress to environmental microbiological results or interesting research summaries and snippets on foodborne disease outbreaks
- run competitions and awareness campaigns and make it fun.

Whatever approach is taken it is important to keep food safety uppermost in the minds of the workers. A short training session on hygiene when they join the company and then nothing more is a sure way to ensure that they are both unaware and unconcerned.

14.5 Learning 4: HACCP plan development and implementation

After training, if there is one other issue that comes up again and again in the book, it is the complexity of HACCP plans and having too many CCPs! This is, of course, linked to a lack of understanding and hence poor quality, or insufficient, education and training. It is unfortunate that many leading HACCP proponents and publications do not clearly understand or agree on the relationship between prerequisite programmes and HACCP, making matters

worse. It must be understood that HACCP is an evolving tool and there is no definitive set of rules. Very few microbiologists, engineers, production or quality managers could claim to be expert in all aspects of food safety management. It may be better to admit this and combine to agree a comprehensive set of guidelines that pulls together HACCP and prerequisite good hygiene practice. We can only highlight this as an issue and, meanwhile, use the lessons gained during the development of HACCP plans. It might be helpful if we used the Seven Principles as subheadings.

14.5.1 Principle 1 – conduct a hazard analysis

Hazard analysis is probably the most difficult principle to carry out correctly for many in both large and small businesses, because of lack of expertise in hazard analysis. Training in HACCP principles and techniques will not equip individuals to be able to undertake hazard analysis unless they have the appropriate understanding of microbiological and chemical hazards. Physical hazards are easier to manage. Experience suggests that people with an awareness of the plant and common sense can often manage such hazards effectively. Two examples that highlight the difficulties are provided by Marthi and by Rudge. Marthi describes how difficult it can be to obtain hazard data from raw material suppliers. He also notes the difficulty of relying on hazard information based on epidemiological data from the USA that is not necessarily relevant or applicable in India. However, as there is no effective disease surveillance programme in India, he and other manufacturers currently have very little choice.

Rudge raises the issue of allergens that are regarded as significant in many developed countries, particularly in the USA and in the UK, but are not seen as a problem in many other countries. How is the plant HACCP team able to decide whether to include them in the hazard analysis if they are not aware that they are a problem? Government level microbiological risk assessment (see later) may help, at least with regard to micro-organisms.

Lee and Suwanrangsi mention government involvement in identifying and providing data on microbiological and other hazards for a specific food sector. An internationally agreed database on hazard prevalence by raw material/product/country or region together with a similarly agreed epidemiological database linking hazards with foodborne disease would be an immense contribution towards improving the quality of hazard analysis in all sectors of the industry, particularly for SMEs. All such data are available. It only needs someone to assemble it in a HACCP user-friendly format!

A point worth mentioning is that made by McAloon in terms of identifying controls for hazards. Instead of putting a control in place, the best method is to remove the hazard by designing it out through process or product redesign. Marthi makes a similar point in discussing the Unilever concept of ‘safety by design’. This approach is often the least used option in contrast to dealing with it in a prerequisite programme or via CCPs.

14.5.2 Principle 2 – determine the critical control points

The biggest problems here seem to be the confusion surrounding the relationship between PRPs and HACCP, and the tendency to include quality as well as safety issues. Killen describes the need to focus on food safety and to reduce CCPs. At Heinz they keep a separate log of spoilage issues that can be dealt with separately and managed by non-critical Control Points (CPs). Suwanrangsi mentions the difficulty that arises through reference to available HACCP models that incorporate both safety and quality control points. It is then necessary for inexperienced companies to try to separate the two in order to deal with the non-safety points through PRPs and the safety-critical points through CCPs. The same author mentions how inexperienced customer auditors sometimes require a manufacturer to include additional, unnecessary non-critical CCPs. This problem was also mentioned by Route, Wallace and Taylor.

McEachern and Lee also highlighted the problem of regulator understanding of HACCP plans. The New Zealand government for example, has, like others, actively sought to develop standards of competency, appropriate qualifications and training programmes for their auditors, but the process takes time. Customer auditors may be even more variable in their competence. It is interesting to compare Kane's positive view of how customer review of supplier HACCP plans can be helpful, where there is an experienced auditor, with the frustration voiced by those asked to add extra and unnecessary 'CCPs' by auditors with less experience.

Perhaps the fact that there is dialogue and disagreement between suppliers, customer and regulator is actually helpful towards increased mutual understanding of effective HACCP implementation. Suwanrangsi mentions that businesses that are most proactive in their culture have the simplest and most effective HACCP plans – they are the ones with well-organised PRPs, a willingness to take full ownership of HACCP and good management skills. The reactive and less confident businesses are those that end up with the over-complex and burdensome systems with far too many CCPs.

Taylor states that good hygiene practice and HACCP have become confused and that there is no clear guidance in the 'standards' on PRPs. The Codex hygiene principles provide the best framework in terms of an international standard but they are lacking in detail and are therefore open to misinterpretation. Taylor provides one diagrammatic representation of the relationship between PRPs and HACCP systems (see Fig. 2.2 on p. 22). Another is provided in Fig. 14.6 (Mortimore and Wallace, 2001).

Management by CCP, CP or PRP depends on understanding the significance of hazards. GMPs can minimise the potential for hazards to occur but they cannot substitute for a CCP as can be seen in an example given by McAloon. Suwanrangsi mentions that additional CCPs might be appropriate as a temporary measure whilst PRPs are being developed. It is unlikely that PRPs would be used to control significant hazards in any case, but adopting this practice, with appropriate labelling, may be useful as a transitional step. It is important to remember that HACCP should be built upon an established framework of

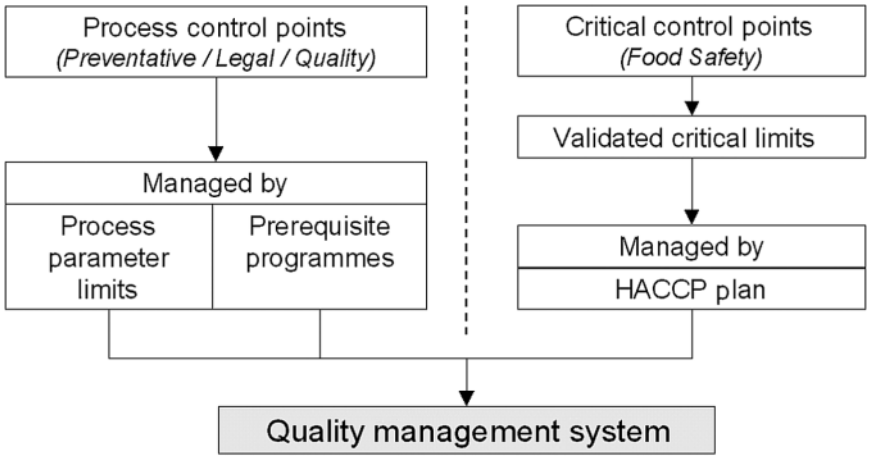


Fig. 14.6 Control point differentiation.

prerequisites/GMPs so the Suwanrangsi approach should be treated with caution, and used only as an interim measure.

14.5.3 Principle 3 – establish critical limits

There are few mentions of this stage in HACCP development in the contributions but it is likely that this can be a problem area, especially for SMEs and less well developed businesses, where relevant expertise is lacking. These businesses may need to use external consultants or other advice to validate limits they establish.

14.5.4 Principle 4 – establish a system to monitor control of the CCP

Barnes and Kane highlighted the issue of CCP monitors who do not understand what they have to do. Route also mentions CCP monitors failing to complete all the paperwork. These problems are substantiated in the UK, for example, by routine inspections. The solution must be found both in training (see earlier), involving line staff in the development of CCPs and supporting documentation, and also through the reinforcement of expectations and systems that can be achieved through regular verification activities.

14.5.5 Principle 5 – establish the corrective action to be taken when monitoring indicates that a particular CCP is not under control

The point that Barnes makes about CCP monitors being on ‘auto-pilot’ and failing to take corrective action when needed has already been discussed, and is potentially a major weakness in a HACCP system. Kane gives a number of

examples where even if a HACCP system was in place, if no corrective action had been taken then a food safety failure would have occurred. It is essential that the procedures are clearly documented and that the CCP monitors are trained to take action. McAloon offers a useful lesson: in writing meaningful corrective action procedures, Cargill set a standard that covered the basic requirements in terms of content. This is an excellent idea, offering rigour and with the ability to involve the operatives in writing CCP procedures within a set framework.

14.5.6 Principle 6 – establish procedures for verification to confirm that the HACCP system is working effectively

Almost all contributors recognised the benefit of audit as part of verification. HACCP principle 6 also encompasses the requirement for validation and verification. The recent ILSI Monograph on Validation and Verification (ILSI, 1999), clarifies the role of the above two activities which, unfortunately are not well explained in the benchmark Codex HACCP text (Codex, 1997b), and consequently are misinterpreted by many people. Part of the role of validation is confirmation that the identified CCP will control the identified hazard. Rudge mentions making sure that the controls will work in practice as opposed to just looking good on paper. Verification is confirmation that the validated CCPs are under control by ensuring that the activities outlined in the HACCP plan are actually carried out! The failure of CCP monitors to monitor CCPs or take corrective action can be addressed through regular verification activity. A key lesson is that having this programme makes the management of CCPs a lot easier and also highlights why it is important to have only the few essential CCPs in the plan. Having to adopt verification on a large number of unnecessary 'CCPs' is what makes HACCP seem to be a burdensome and irrelevant exercise.

Verification activities include a review of critical records, consumer complaint monitoring, calibration of critical measuring devices and audit. McAloon sees self-audit as the key to long-term success. This is a view also expressed by others. Some ideas that might be of help include:

- Internal auditors require a high degree of knowledge and experience. Ideally this should have been gained from working directly on the process being audited.
- Auditors must be independent of the process being audited, but it still works well if they are drawn from production personnel involved in the process.
- The staunchest critic of the business is the business itself – try to ensure that the audit function is viewed positively and acknowledge that the intention is to improve the system.
- A regular (e.g. monthly) audit, if done well, will not only identify deficiencies but can also encourage staff to suggest improvements and ways of simplifying procedures.

14.5.7 Principle 7 – establish documentation concerning all procedures and records appropriate to these principles and their application

Given that in some countries (e.g. the UK) there has been a legal requirement for a HACCP approach but with no requirement for documentation, it was surprising that there were only a few comments in this area. Again, the main problem is correctly identifying the true CCPs that enable documentation to be made manageable. Some ideas were offered by the contributors in dealing with the problems they had observed or encountered:

- How to integrate documentation without adding complexity (McAloon); the answer lies in using existing paperwork and forms. Auditing helped to highlight where these needed modification and an operator can be asked to help in the redesign.
- Management of critical records, especially those related to non-conformity (Rudge); use of simple databases is recommended as a solution where paper systems become unwieldy. The lesson here is to emphasise that the HACCP system is there to work for the benefit of the business and to add control. Using a database means that not only can records be organised but trends can more easily be analysed, which is better for longer-term preventative actions.
- Documentation is most easily managed where there is an existing quality management system. Taylor points out that many of the SMEs do not have the benefit of such a framework to build on.

In summary, the main problems that have been highlighted related to over-complex HACCP plans with too many CCPs. This is, largely, a result of misunderstanding and confusion between PRPs (for CPs) and CCPs. However, there were some more surprising discoveries made by McAloon during the development of the HACCP programmes at Cargill.

- Eighty per cent of the pieces needed for HACCP were there already – they just needed organising. This is probably the case for many large and small businesses.
- Line workers showed commitment earlier than their supervisors.
- Challenging the status quo is good practice.

14.6 Learning 5: HACCP maintenance activities

Kane observes that, from a retailer perspective, it is clear that failure to maintain HACCP systems can be a major weakness. As an industry there are numerous examples where food safety incidents occurred as a result of a process change (e.g. a food poisoning outbreak caused by hazelnut yoghurt in the UK – where a reduced sugar formula was not assessed for revised thermal process requirements).

14.6.1 How to keep momentum going

After all the effort of getting a HACCP plan developed it can be difficult to keep the motivation and momentum going, particularly once the next major project comes along. This is a key issue to bear in mind when starting a HACCP programme. To use the HACCP programme to create a sustained and proactive commitment to food safety takes a lot of thought and planning. McAloon describes how recognition tools can be used to good effect. Simple and inexpensive rewards can make a big difference to the ongoing commitment within the workforce. Marthi describes the workforce as becoming 'empowered' which, once achieved, should not be undermined.

14.6.2 People issues – staff turnover, ongoing training, absenteeism

It is often easier to train new people into the business than to train your existing employees. In an established HACCP system there will usually be two guaranteed training needs as a result of staff turnover and the legacy of veteran employees. It is important to recognise the differences between the two groups and to plan a different approach to ensure that both groups are trained properly.

Training new employees as a result of staff turnover

This is a problem not just for large companies but also for SMEs who often employ casual workers, particularly in the foodservice sector. McAloon really drives home this point with the example of one plant employing 2,000 people speaking four different languages, where there was 20% annual turnover. He also shared the fact that after twelve months of the HACCP programme being operational the turnover, particularly at supervisor level, was having an impact and could be seen in their internal audit scores. There are a number of solutions to consider:

- An orientation programme for HACCP activities can be set up. As Wallace points out this can also be done for hygiene training.
- Ensure that key workers are identified (CCP monitors, supervisors) and that where job descriptions exist, their special food safety responsibilities and training needs upon recruitment are clearly identified.

Refresher training for existing employees

McAloon suggests short bursts of training (15–20 minutes each month) which can be more easily incorporated into the regular work pattern than half-day or full-day courses. Absenteeism is another problem for many businesses and it is necessary to build cover for key workers into the system rather than to assume that they will never take a vacation or get sick.

14.6.3 Larger company issues

There are a couple of lessons that are mainly related to larger multinational businesses.

What happens to the corporate HACCP support team once the HACCP project is completed?
McAloon relates how important it is to agree roles and responsibilities (for instance between the plant and headquarters) and communicate what these are. While the plant team usually maintains the HACCP system, the HQ team may be used to develop a library of resource and training materials. Alternatively they may assume responsibility for tracking and communicating information on such things as emerging hazards, or they may be used in a verification role – reviewing HACCP plans and auditing. These are important and useful activities and it may be possible for smaller businesses that do not have this resource to get some of these services via other means, e.g. industry trade associations, regional food safety clubs, food research associations and reputable consultants.

How should acquisitions and co-packers be incorporated into the HACCP system?
Again, this is a point raised by McAloon, but one that will be recognised by anyone working in a larger company. The key lesson is not changing anything in a rush or automatically. Instead one should acknowledge the experience of the people working for the newly acquired company or co-packer. It is important to understand their point of view before starting the HACCP development project in much the same way as described earlier, i.e. baseline audit, training plan, project plan, etc.

14.6.4 Process improvements

Killen and McAloon shared their experiences in this area as it is obvious that, for progressive companies, HACCP is seen as an opportunity for improvement. HACCP helps a business to understand its processes better, introduces a range of improvements, and acts as a platform from which to go on to use more complex process improvement tools.

14.6.5 Regulatory assessment

There are a number of approaches described in the texts which can be compared to the very good recent WHO/FAO Consultation on the role of regulatory bodies in the assessment of HACCP (WHO/FAO, 1998): Guidance on Regulatory Assessment of HACCP). While HACCP verification (Principle 6) is clearly a key responsibility of the food business, regulatory assessment provides both an important service to the company, in providing an objective assessment of a HACCP system, and serves the consumer by virtue of the fact that the regulator is acting on their behalf to ensure compliance with food safety standards. A number of lessons emerged from the regulatory contributions:

- Checklists can be a useful tool (Lee) but they can never be all inclusive and the assessor needs to apply experience and judgement.

- The relationship between an inspector and a food company can be seen as adversarial (McEachern). In the Canadian programme, as in most if not all countries, regulatory inspections are not done at the invitation of the company.
- Regulatory assessment can be used to promote industry ownership of and competence in HACCP, i.e. as a confirmation that all of the required components are in place.

What comes across with all the regulatory contributions, and to some extent Kane's retailer perspective, is how important it is for industry and regulator to work together for the benefit of the consumer whom they both serve. The changing roles and different approaches taken by regulators are interesting and their views are helpful to all businesses who have to operate further back in the supply chain to ensure the safety of raw materials. If the food safety regulation programme is well developed, is there any need for businesses to audit their own suppliers or is this a duplication of effort?

14.7 Where next for businesses with HACCP systems?

Once they have set up a HACCP system, where do businesses go next? How should they develop established HACCP systems? Are there opportunities highlighted in the individual contributions, or are there changes already happening within the industry and/or regulatory environment that will facilitate change? Certainly we have learned that a number of businesses have used HACCP as a springboard for continuous improvement. McAloon, Rudge and Killen take the view that it can lead very logically into process analysis and subsequent improvements. Kane suggests that the concept can be extended into other aspects of food safety such as malicious contamination. The approach can also be applied to non-food areas – McAloon mentions human safety. Alternatively, the energy and cultural change that may have developed through the use of the HACCP approach can be harnessed to drive other quality initiatives. Can HACCP be used further in other ways, as a competitive tool to drive broader business benefits? Perhaps the opportunities may include:

- setting up measuring systems to monitor overall HACCP plan performance and to identify improvements to the system, for example in CCP compliance
- referring to published benefits (for example cost saving, waste reduction, process improvement, sanitation downtime reduction) and setting objectives for achievement of measurable costed improvements
- extending the skills developed to other non-food safety or non-hazardous areas of the business, for example the critical sensory attributes of the product, or identifying the critical process steps in a commercial transaction process
- ensuring access to emerging hazard data and to being proactive in making sure that the HACCP system is reviewed against this data

- setting up a system to ensure that equipment, process or recipe changes are captured and incorporated into the system
- reviewing regulatory and consumer assessment frequency based on good performance
- reducing the QA costs associated with the current practice of end product testing by relying instead on validated HACCP plans to provide real 'on line' process and product control. Apart from providing the manufacturer with an increased confidence in the safety of the product (the *raison d'être* for HACCP), this has a number of additional benefits, including the quicker release of products to market with a reduced requirement for storage of product 'on hold'.

14.8 Summary

It is time to pick up some of the issues first raised in the introduction. Is the concept of HACCP flawed in some way? A number of issues have been raised in this chapter which suggest the need to reassess HACCP principles as they have subsequently developed and been codified. These issues are:

- continuing confusion over the relationship of HACCP to prerequisite programmes (PRPs)
- the early development of what can be seen as a more elaborate 'large industry' HACCP model, partly by Codex and other experts, and partly by the early pioneers of HACCP systems, who were mostly large companies.

There is now a move to 'second-generation' HACCP models allowing greater flexibility and putting more emphasis on prior development of effective PRPs with simpler HACCP systems focusing much more effectively on the key hazards. This approach is seen as more effective for SMEs and businesses in other parts of the food chain than manufacturing, for example food service operations. It suggests the continued need to reassess what HACCP principles really mean in practice. What does the practical experience described in this book tell us about those principles?

- It confirms that, when a business understands those principles well and has the commitment and resources to carry them out, HACCP works. A number of contributors describe the major contribution of HACCP not only to food safety but to the overall competitiveness of the business.
- Often, 80 per cent of the requirements for a HACCP system are already present; they simply need to be identified and integrated into a HACCP framework.
- HACCP is a 'minimal' system allowing businesses to make the most of their existing systems and focus resources more effectively on the critical safety areas.
- Because HACCP is a minimal system, it is clear that HACCP is not just the province of the multinational but is equally applicable to SMEs too. Indeed,

SMEs have a number of advantages over larger companies in HACCP implementation, including simpler processes to analyse, simpler lines of communication, greater flexibility and speed.
- If implemented effectively, HACCP can bring significant commercial benefits, not just in avoiding the legal and financial implications of a serious food safety incident, but in reducing costs and improving productivity. It is also empowering, allowing a business to understand and improve its processes, and producing more proactive employees able to contribute more to the business.

If HACCP can work so well, why do so many businesses have problems with it? The contributions in this book suggest the main problems are:

- Wrong perceptions of the value and complexity of HACCP implementation, often linked to low motivation in embracing HACCP and its benefits; poor motivation can sometimes be linked to difficulties in measuring and quantifying the benefits of a HACCP system.
- The need for culture change within an organisation; HACCP requires a proactive culture throughout the organisation, from senior management to line operators, involving greater responsibility for identifying and controlling hazards. Some organisations have a more traditional and hierarchical culture in which managers respond reactively to food safety issues and line staff are given little opportunity to show initiative.
- Misunderstanding of the HACCP concept, for example in incorrectly identifying CCPs.
- Lack of appropriate expertise in such areas as hazard analysis.

These problems clearly affect not only large and small companies in the food industry but also regulators and customers (such as retailers) who set and enforce standards, and provide advice and support. Such fundamental problems then lead to a familiar series of HACCP system failures:

- a misunderstanding of the role of PRPs, the hazards to look for and the nature CCPs, resulting in over-complex and unmanageable HACCP systems
- ineffective monitoring and corrective action, due to failures in organisational culture, poor training and verification procedures
- poor documentation, as a result of over-complex HACCP systems
- poor validation, especially for SMEs lacking the relevant expertise on hazards.

Against this background of common systems failures, there are distinctive problems across countries, including language, cultural differences and quality of infrastructure. The core problems which produce HACCP systems failure have, in varying degrees, affected the contributors to this book, and they provide a range of proven, practical solutions to them which have been summarised in this chapter. The contributors demonstrate that, although there are challenges to HACCP implementation, they can be met and the rewards are well worth the investment. One common theme, especially for SMEs, is the importance of

appropriate support from others, whether regulators, trade bodies or customers such as retailers. As well as setting and enforcing standards, governments, for example, are developing innovative ways of promoting understanding of HACCP, developing training and other resources, and providing expertise to fill such skills gaps as hazard analysis. Partnership, whether within particular sectors of the food industry, between customers and suppliers, or between government and industry will be an increasingly important factor in making HACCP implementation more effective and widespread in the food industry.

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